3 Network Theorems

INTRODUCTION

Network theorems are very useful for simplifying complicated electric circuits, which consist of several sources, nodes and meshes where the response in a single element is desired. These theorems are useful for both dc and ac circuits. The objective of this chapter is to introduce different theorems.

1) Superposition Theorem
2) Thevenin's Theorem
3) Norton's Theorem
4) Maximum Power Transfer Theorem (MPT theorem)
5) Reciprocity Theorem

Definitions

Linear circuit: "It is defined as a circuit whose output is linearly related to its input".

LINEARITY PROPERTY

"The linear property of an element is used to describe a linear relationship between the cause and the effect".

An element is said to be linear only if it satisfies the two properties:

- Additivity (superposition) property
- Homogeneity (scaling) property

Additivity property: "If an input $x_1(t)$ produces an output $y_1(t)$ and another input $x_2(t)$ produces output $y_2(t)$, then when the sum of $x_1(t)$ and $x_2(t)$ is given as an input, produces an output $y_1(t) + y_2(t)$".

$$x_1(t) \rightarrow y_1(t)$$
$$x_2(t) \rightarrow y_2(t)$$
then, $$x_1(t) + x_2(t) \rightarrow y_1(t) + y_2(t)$$

Homogeneity property: "In homogeneity property, if an input (excitation) is multiplied by a constant a, the output (response) will also be multiplied by the same constant".

- The two properties of linearity and additivity that defines a linear system that can be combined into a single equation as "for any linear resistive circuit, the output voltage or current is denoted by the variable, Y is related to linear independent sources (inputs)", i.e.,

$$Y = a_1x_1 + a_2x_2 + \ldots\ldots + a_nx_n$$

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Among the non-linear elements, such as transistors, tend to linearity when small AC signals are applied to them.

Where x_1, x_2, ... x_n are the voltage values or the current values of the independent source present in the circuit and a_1, a_2.... a_n are the arbitrary constants.

For the linearity principle, consider a linear circuit shown in figure 3.1. It does not contain independent sources. The circuit is excited by the source voltage (V_s), which serves as an input. The circuit is connected to a load resistance R, and the current 'i' through the R is taken as an output, as shown in the figure.

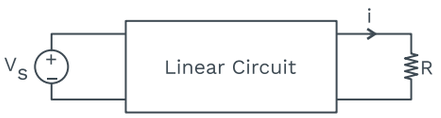


Fig. 3.1 A Linear Circuit with an Input Voltage V_s and an Output Current I

Note: When a current i_1 flows through a resistor R, the power in the circuit is $P_1 = Ri_1^2$ and when a current i_2 flows through R, then the power is $P_2 = R \times i_2^2$. If the sum of both the currents $i_1 + i_2$ flows through resistance R then the power absorbed is $P_3 = R \times (i_1 + i_2)^2 \neq P_1 + P_2$. Thus, the power is considered as a non-linear parameter.

SUPERPOSITION THEOREM

"For a linear, bilateral network, which consists of a number of energy sources, resistances etc., then the effect produced by an element, when all the sources in the circuit act at an instant, is equal to the summation of the effect produced by the same element when the sources are considered individually".

- This theorem can only be applied to a linear, bilateral network.
- We can find linear parameters of the network such as voltage and current by means of superposition but not non-linear parameters such as power ($\propto i^2$).

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The superposition theorem is not limited to the circuit analysis but it is applicable in many fields where the cause and effect bears a linear relationship to one another.

Points to be considered before applying super position theorem is:

- All the independent voltage sources are short-circuited or replaced by their internal impedances.
- The independent current sources which are present in the circuit, are open-circuited, or are replaced by their internal impedances.

Definitions

Superposition Theorem: "It states that the response of a network with multiple sources will be the sum of the responses due to each source".

- All dependent voltage and current sources remain as it is. These sources are neither short circuits nor open circuits.

Features of superposition theorem:

- It is applicable to only linear networks i.e., the network consisting of R, L, C, transformer and linear dependent sources as elements. These elements are either time-variant or time-invariant.
- This theorem applies to both active and passive networks.
- This theorem applies to the network that contains initial conditions.
- This theorem is preferable applicable to those networks which contain more than one source.

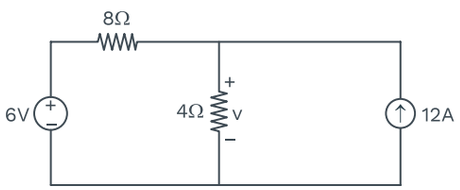
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Limitations of the superposition theorem are mentioned below:

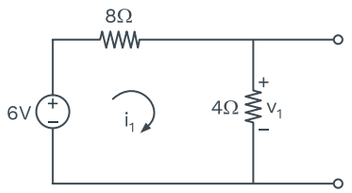
- It is not applicable to the network that consists of non-linear elements such as p-n junction diode, transistors, semiconductor devices etc. and unilateral elements such as diodes, BJT's etc.
- It is not applicable to the calculation of power as power is a non-linear parameter.

Note:
The superposition theorem is not valid for non-linear networks and non-linear parameters like power.

Example 1:
Calculate the voltage across 4Ω resistor by using the superposition theorem.



Sol:
Consider a 6V voltage source and deactivate the current source (i.e., the current source is open circuit), then the circuit is redrawn as shown below.



On applying mesh analysis in the network,

Rack Your Brain

Why superposition theorem is not applicable to unilateral elements?

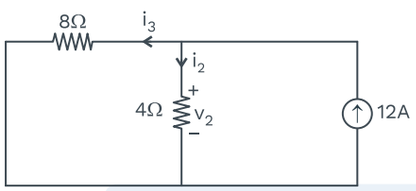
$$-6 + 8i_1 + 4i_1 = 0$$

$$i_1 = \frac{6}{12} = 0.5A$$

Since, $v_1 = 4 \times i_1$

$v_1 = 2$ V

Now considering the current source and voltage source are short-circuited, the circuit is redrawn, as shown below.



By using the current division, we get:

$$i_2 = \frac{12 \times 8}{8 + 4} = 8A$$

Since, $v_2 = 4 \times i_2$

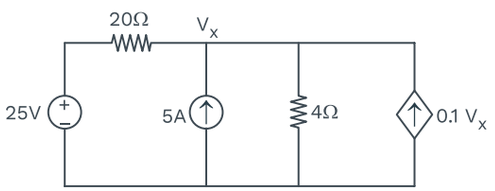
$v_2 = 4 \times 8 = 32$ V

To find voltage across 4Ω resistance,

$$v = v_1 + v_2 = 2 + 32 = 34 \text{ V}$$

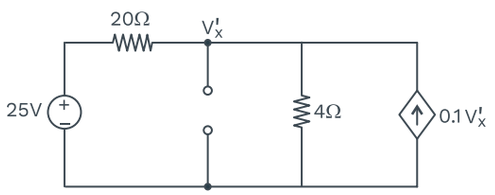
Example 2:

By using the superposition theorem calculate the voltage shown in the circuit.



Sol:

Consider a 25V voltage source, and the current source (5A) will be open circuited; the circuit is redrawn as shown below.



Previous Years' Question

The superposition theorem is not applicable to networks containing:

a) Non-linear elements
b) Dependent voltage sources
c) Dependent current sources
d) Transformers

Sol: a)

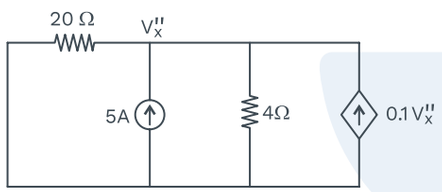
Applying nodal analysis at V_x', we get:

$$\frac{V_x' - 25}{20} + \frac{V_x'}{4} - 0.1V_x' = 0$$

$$\frac{6V_x'}{20} - 0.1V_x' = \frac{25}{20}$$

$$V_x' = 6.25 \text{ V}$$

Now consider 5A current source and the voltage the source will be short-circuited and the circuit is redrawn as shown below.



On applying the nodal analysis at V_x'', we get:

$$\frac{V_x''}{20} - 5 + \frac{V_x''}{4} - 0.1V_x'' = 0$$

$$0.2V_x'' = 5$$

$$V_x'' = 25 \text{ V}$$

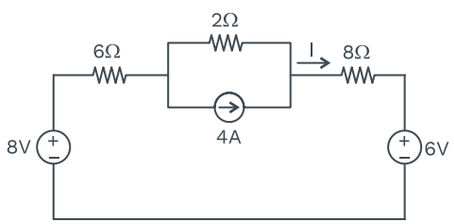
$$\therefore \quad V_x = V_x' + V_x''$$

$$V_x = 6.25 + 25$$

$$V_x = 31.25 \text{ V}$$

Example 3:

Obtain the value of current I in the given circuit, by using the superposition theorem.



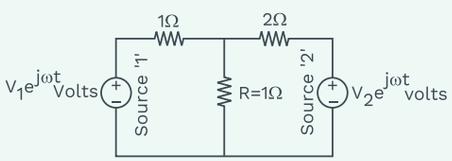
Sol:

In the given circuit three sources are present,

Consider, $I = I_1 + I_2 + I_3$

Previous Years' Question

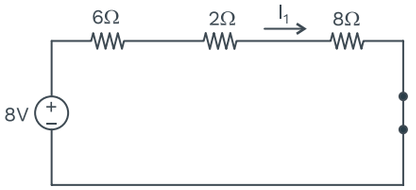
In the circuit shown in the figure, the power dissipated in the resistor R is 1W. When only source '1' is present and '2' is replaced by a short circuit. The power dissipated in the same resistor R is 4W. When only Source '2' is present and '1' is replaced by short-circuit. When both the sources are present, the power dissipated in R, will be:



a) 1 Watts **b)** 3 Watts

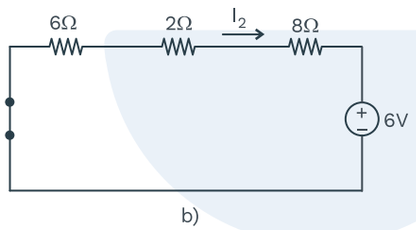
c) 4 Watts **d)** 5 Watts

For the circuit shown in figure a), the current I_1 is:



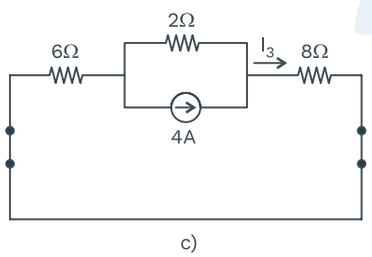
$$I_1 = \frac{8}{16} = 0.5\,A$$

For the circuit shown in figure b), the current I_2 is:



$$I_2 = \frac{-6}{16} = -0.375\,A$$

For the circuit shown in figure c), the current I_3 is:



$$I_3 = 4 \times \frac{2}{2+14} = \frac{8}{16} = 0.5\,A$$

$I = I_1 + I_2 + I_3 = 0.5 - 0.375 + 0.5$

$I = 0.625\ A = 625\ mA$

Definitions

Source Transformation: The transformation of the source is a the technique used for the circuit simplification to modify the complex circuit by transforming the independent voltage source to the independent current source and vice-versa.

SOURCE TRANSFORMATION

A source transformation is defined as a method of simplifying circuits in which a voltage source (V_s) in series with a resistor R is replaced by an equivalent current source (I_s) in parallel with a resistor R, or vice-versa.

Note: Source transformations are implemented by using the Thevenin's theorem and Norton's theorem.

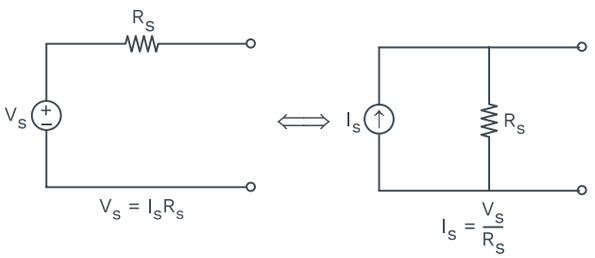
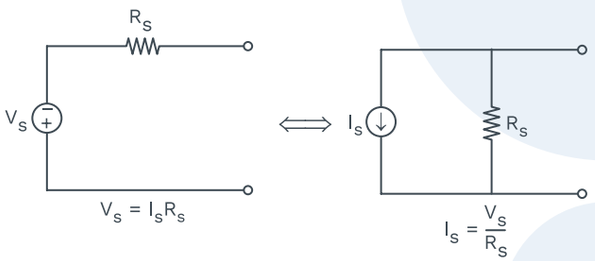


Fig. 3.2 Source Transformation of Independent Sources

Note:

The head of the current source arrow corresponds to the positive terminal of the voltage source (as shown below).



- The conversion of sources will be equivalent at their external terminal only i.e. the voltage and the current relationship at their external terminals remains the same.
- Source transformation cannot be applied to an ideal voltage source since for an ideal voltage source $R_s = 0$. So, equivalent current source value $(I_s) = \frac{V_s}{R_s} \to \infty$. Similarly, it cannot be applied to the ideal current source, as source resistance to an ideal current source will be infinite ($R_s = \infty$), so the equivalent voltage source value will not be finite.

SOURCE TRANSFORMATION FOR DEPENDENT SOURCE

Source transformation can also be applied to the dependent sources in the same manner as independent sources. Figure 3.3 shown below is the source transformation of a dependent source.

Rack Your Brain

Why is the current source in the source transformation always in parallel with the elements?

Grey Matter Alert!

The Thevenin equivalent circuits and the Norton equivalent circuits are related by the source transformation.

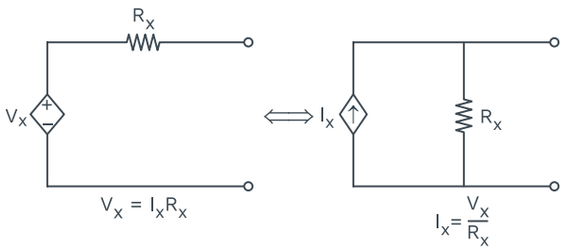


Fig. 3.3 Source Transformation of Dependent Sources

Rack Your Brain

What are the applications of source transformation?

THEVENIN'S THEOREM

"A linear network consisting of a number of voltage sources and resistances can be replaced by an equivalent network having the single voltage source called "Thevenin's voltage" (V_{Th}) and a single resistance called as "Thevenin's resistance" (R_{Th})".

According to the theorem, a linear circuit as shown in figure 3.4 a) is replaced by the equivalent circuit, as represented in figure 3.4 b).

The circuit connected to the left of the terminals a and b in figure 3.4 b) is called the Thevenin equivalent circuit.

Grey Matter Alert!

Thevenin's theorem was introduced by the German scientist, Hermann von Helmholtz in the year 1853 and by Leon Charles Thevenin in the year 1883. Thevenin (1857-1926) was an electrical engineer in France's, national Postes et telegraphs telecommunications organization.

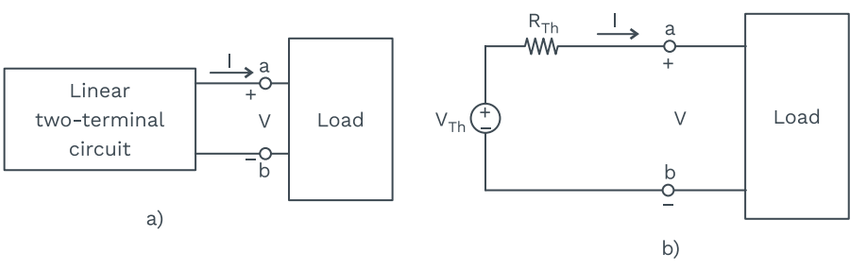


Fig. 3.4 a) Linear Two-Terminal Circuit b) Thevenin Equivalent Circuit

"It states that a linear, bilateral, two-terminal networks can be replaced by an equivalent circuit consisting of the voltage source (V_{Th}), in series with a resistance (R_{Th}) (or) impedance (Z_{Th}), where V_{Th} is an open circuit voltage between the terminals and R_{Th} and Z_{Th} is an equivalent resistance and impedance respectively across the terminals when the independent sources in the circuit are absent".

THEVENIN VOLTAGE (V_{Th})

"If the terminals a and b are made open-circuited (by removing the load), thus, no current flows, so the voltage across the terminals a and b as shown in figure 3.4a) is equal to the voltage source V_{Th} figure 3.4b), since the two circuits are equivalent to each other".

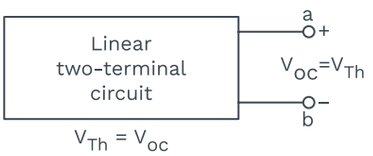


Fig. 3.5 a) Finding Thevenin's Voltage V_{Th}

$$V_{Th} = V_{OC}$$

Where V_{Th} = Thevenin's voltage
R_{Th} = Thevenin's resistance

Note:

If a circuit contains a dependent source only means that there is no independent source present in the network, then the open-circuit voltage or Thevenin voltage will be zero.

THEVENIN RESISTANCE (R_{Th})

- With the load disconnected, the terminals a and b are open-circuited, the voltage source is replaced by the short-circuit and the current source is replaced by the open-circuits.
- The input resistance (equivalent resistance) of the dead circuit as shown in figure 3.4 a) is equal to the Thevenin's resistance (R_{Th}) in figure 3.4b) since both the circuits are equivalent.
- Thus, the Thevenin's resistance (R_{Th}) is equal to the input resistance at the terminals a and b when the independent sources are turned off, as shown below.

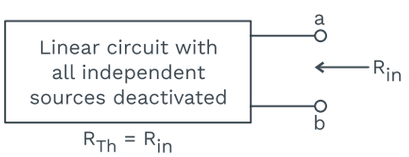


Fig. 3.5 b) Finding Thevenin's Resistance R_{Th}

Methods for finding R_{Th}:

Rack Your Brain

Why is Thevenin's theorem not applicable to non-linear circuits?

Definitions

Thevenin's theorem: "Thevenin's theorem states that a network containing sources and resistors (linear bilateral and active) can be represented as a voltage source in series with a resistance across a pair of terminals a and b".

Grey Matter Alert!

Thevenin theorem does not apply to the network containing:
a) Non-linear element
b) Unilateral element

Case 1:
Circuit with Independent sources only.
This case is useful for finding R_{Th} by the direct method. In this, the voltage source will be short-circuited and the current source will be open-circuited and then find R_{Th} as shown in figure 3.5b).

Case 2:
- Circuit with dependent sources.
- A voltage source V_0 is applied at terminal a and b and then obtain the resulting current i_0 (in the circuit shown below).



Fig. 3.6 Finding R_{Th} when the Circuit has a Dependent Source

Then $R_{Th} = \frac{V_0}{i_0}$

Since the dependent source cannot be expressed in terms of resistance, thus, determining R_{Th} is not possible directly instead, we use ohm's law where $R_{Th} = \frac{V_{oc}}{I_{sc}}$ at the desired terminals.

Finding R_{th} using short-circuit current:

Connect a short circuit between the terminals a and b in figure 3.4 a). The independent sources are not set to zero as we are required to find short-circuit current. Thus obtain the short-circuit I_{sc} through the terminal a and b. Then the Thevenin resistance is given as:

$$R_{Th} = \frac{V_{Th}}{I_{sc}}.$$

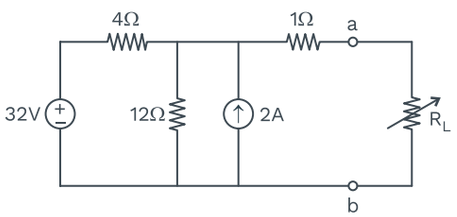
Note:
If the value of R_{Th} is negative, it indicates that the circuit is supplying the power i.e. (v = –iR).

Rack Your Brain

Is it possible to convert an ideal voltage source into an ideal current source in a circuit?

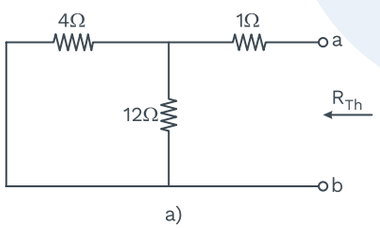
Example 4:

For the circuit shown below, calculate the Thevenin equivalent across the terminals a and b. Then obtain the current through the load resistance $R_L = 11\Omega$.



Sol:

Thevenin resistance R_{Th} can be obtained by replacing the voltage source with a short-circuit and replacing the current source with an open-circuit, as shown below in figure a).



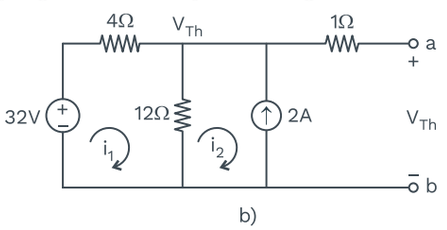
a)

$$R_{Th} = (4||12) + 1$$

$$= \frac{4 \times 12}{4 + 12} + 1 = 4\Omega$$

To obtain the Thevenin voltage V_{Th}, consider the circuit in figure b).

By applying mesh analysis, we get:



b)

$-32 + 4\, i_1 + 12(i_1 - i_2) = 0$; $i_2 = -2A$

By solving the above equation, we get:

$i_1 = 0.5$ A

Thus, $V_{Th} = 12 \times (i_1 - i_2) = 12 \times (0.5 + 2) = 30$ V

So, the Thevenin equivalent circuit as required in the question is shown below:

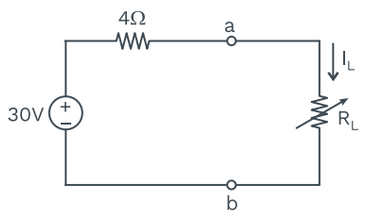
Grey Matter Alert!

In a circuit, if $R_{Th} = \infty$, then its Thevenin equivalent circuit, is not possible. We can only find its Norton equivalent circuit.

In a circuit if $R_{Th} = 0$, then its Norton equivalent circuit is not possible. We can only find its Thevenin equivalent circuit.

Rack Your Brain

What is the use of Thevenin's theorem?

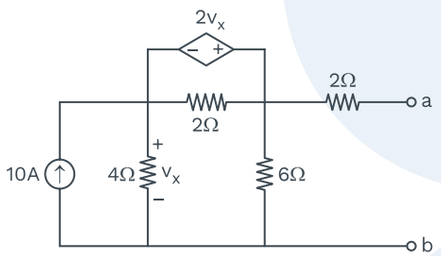


$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{30}{4 + R_L}$$

When $R_L = 11\Omega$, $I_L = \frac{30}{15} = 2A$

Example 5:

For the given circuit, obtain the Thevenin equivalent at the terminals.



Sol:

To obtain R_{Th}, the independent sources are set to zero without changing the dependent source. Because of the dependent source, the network is excited with a voltage source v_0 connected at the terminals.

To ease the calculation, $v_0 = 1V$. Then calculate the current i_0 in the linear circuit and then obtain $R_{Th} = \frac{1}{i_0}$.

Now apply mesh analysis in loop-1 in figure a), we get-

$$-2v_x + 2(i_1 - i_2) = 0$$

$$v_x = i_1 - i_2$$

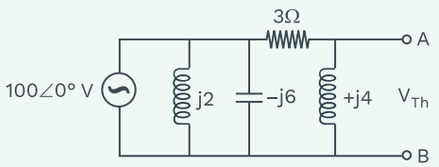
But, $v_x = -4i_2$

$$i_1 - i_2 = -4i_2$$

$$i_1 = -3i_2$$

Previous Years' Question

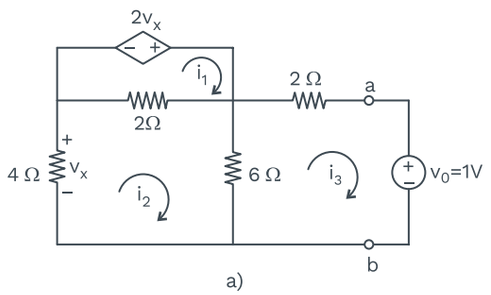
The Thevenin equivalent voltage V_{Th} appears between the two terminals A and B in the network shown below are given by:



a) j16 (3 – j4)V **b)** j16(3 + j4)V

c) 16 (3 + j4)V **d)** 16 (3 – j4)V

Sol: a)



a)

Now applying KVL in loop-2 and loop-3 and we get:

$$4\,i_2 + 2(i_2 - i_1) + 6(i_2 - i_3) = 0$$

$$6(i_3 - i_2) + 2\,i_3 + 1 = 0$$

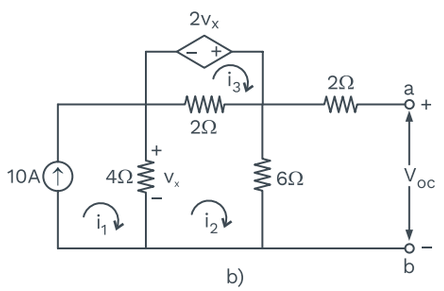
On solving the above equations, we get:

$$i_3 = -\frac{1}{6}\text{A}$$

But $$i_0 = -i_3 = \frac{1}{6}\text{A}$$

Hence, $$R_{Th} = \frac{v_0}{i_0} = \frac{1}{\frac{1}{6}} = 6\Omega$$

To find V_{Th}, calculate V_{oc} in figure b).



b)

Now apply mesh analysis in figure b), we get:

$$i_1 = 10\text{A} \qquad \text{...(i)}$$

$$-2\,v_x + 2(i_3 - i_2) = 0$$

$$V_x = i_3 - i_2$$

But, $4(i_1 - i_2) = v_x$

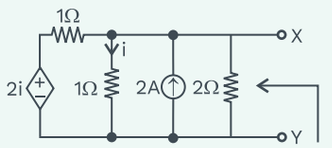
$$4(i_1 - i_2) = i_3 - i_2$$

$$4i_1 - 3i_2 - i_3 = 0 \qquad \text{...(ii)}$$

$$-4i_1 + 12i_2 - 2i_3 = 0 \qquad \text{...(iii)}$$

Previous Years' Question

For the given circuit, determine the Thevenin voltage and the Thevenin resistance across the terminal X and Y.



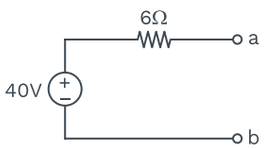
a) $\frac{4}{3}\text{V}, 2\Omega$ **b)** $4\text{V}, \frac{2}{3}\Omega$

c) $\frac{4}{3}\text{V}, \frac{2}{3}\Omega$ **d)** $4\text{V}, 2\Omega$

By solving equations i), ii), and iii), we get:

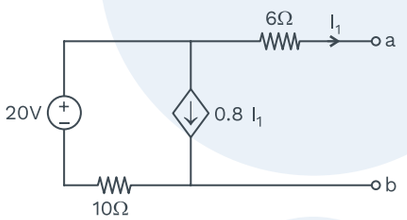
$$i_2 = \frac{20}{3} A$$

Hence, $V_{Th} = V_{oc} = 6i_2 = 6 \cdot \frac{20}{3} = 40V$



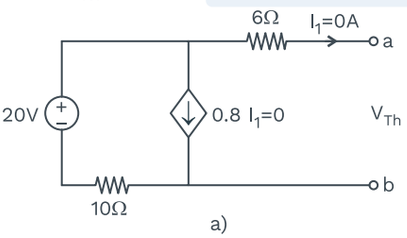
Example 6:

Consider the circuit shown below, and calculate the Thevenin equivalent circuit across terminals a and b.



Sol:

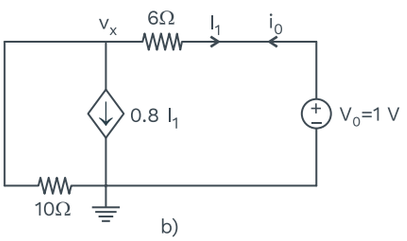
To find Thevenin's voltage (V_{Th}), consider the circuit in figure a).



a)

V_{Th} = 20V

To obtain Thevenin's resistance (R_{Th}), the independent source is set to zero without affecting the dependent source in figure b). We may set V_0 = 1 V to simplify the calculation since the circuit is linear.



b)

Applying nodal analysis, we get:

$$\frac{v_x}{10} + 0.8I_1 + I_1 = 0$$

$$v_x + 18\ I_1 = 0$$

As, $I_1 = \frac{v_x - 1}{6}$

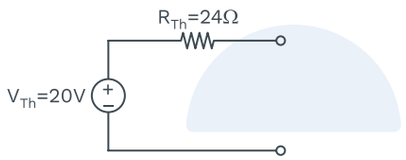
$$v_x + 18\left(\frac{v_x - 1}{6}\right) = 0$$

$$v_x = \frac{3}{4}$$

$$I_1 = \frac{\frac{3}{4} - 1}{6} = -\frac{1}{24}$$

$$i_0 = -I_1 = \frac{1}{24}$$

$$R_{Th} = \frac{v_0}{i_0} = \frac{1}{\frac{1}{24}} = 24\Omega$$



Note:

i) We can never connect an independent voltage source across a dependent voltage source.

ii) We can never connect an independent current source across a dependent current source.

NORTON'S THEOREM

"This theorem states that any linear, bilateral, two-terminal networks is replaced by an equivalent circuit consisting of the current source (I_{SC}) in parallel with resistance (R_N) or impedance (Z_N), where I_{SC} is the short circuit current between the terminals and R_N or Z_N is equivalent resistance or impedance at the terminals, when the independent sources are disconnected".

Norton is the dual of Thevenin's theorem.

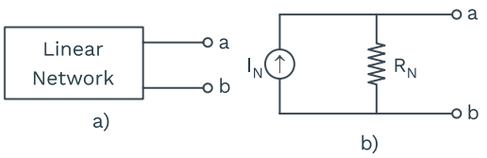


Fig. 3.7 a) Linear Network
b) Norton Equivalent Circuit

I_N = Norton's current
= Short circuit current across a and b

R_N = Norton's resistance

- Norton's theorem is similar to Thevenin's theorem, but it uses the current source in place of the voltage source.
- Thevenin and Norton can be obtained from each other by source transformation.

NORTON'S CURRENT (I_N)

To find Norton's current I_{SC}, connect a short circuit between terminals a and b. In this method, the independent sources are not set to zero, as short-circuit current is required. Now obtain the short-circuit I_{SC} through the terminal a and b.

- The dependent and the independent sources are treated similar way as in Thevenin's theorem.

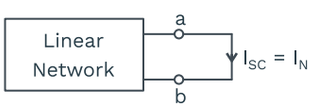


Fig. 3.8 Finding Norton's Current

Norton's current = short circuit current i.e. $I_N = I_{SC}$ as shown in figure 3.8.

Note: If the circuit contains dependent sources only, i.e. there is no independent source present in the network, then its short-circuit current or Norton current will be zero.

Rack Your Brain

Can the Thevenin theorem be used with AC circuits?

Grey Matter Alert!

Norton's theorem was introduced by Siemens and Halske in 1926 and researcher Hans Ferdinand Mayer (1895-1980) and Edward Lawry Norton (1898-1983).

Definitions

Norton's theorem: "A linear network consisting of a number of voltage sources and resistances can be replaced by an equivalent network having a single current source (I_N) and a single resistance (R_N)".

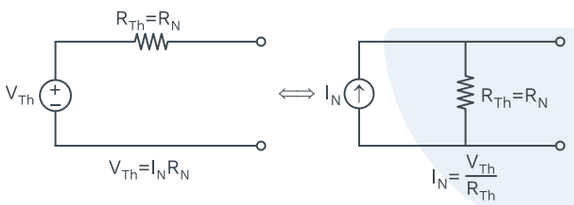
NORTON'S RESISTANCE (R_N)

"Norton resistance is the input or the equivalent resistance seen at load terminal, when all independent sources are set to zero i.e. voltage source is replaced by a short circuit, and the current source is replaced by an open circuit".

Thus, Norton's resistance is the same as Thevenin's Resistance.

Note:

Since the Thevenin equivalent replaces the entire network with a voltage source in series with resistance, the same network can also be represented as a current source in parallel with resistance and hence, the Thevenin and Norton's theorem is related to source transformation.



$\therefore \quad R_N = R_{Th}$

Keep in mind:

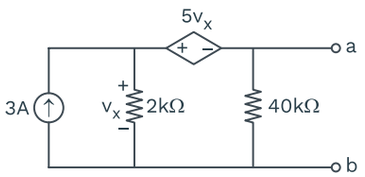
- If a circuit has $R_{Th} = \infty$, then its Thevenin equivalent is not possible. We can only find its Norton's equivalent.
- If a circuit has $R_{Th} = 0$, then its Norton's equivalent is not possible. We can only find its Thevenin equivalent.

Rack Your Brain

What is the use of Norton's theorem?

Example 7:

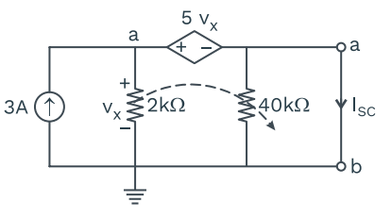
In the circuit shown, the reading of the practical ammeter resistance 5 kΩ connected across terminals a-b is-



Sol:

Step 1:

Compute I_{SC}



Applying nodal analysis at point a:

$$\frac{v_x}{2\times10^3} - 3 + I_{SC} = 0$$

$$I_{SC} = 3 - \frac{v_x}{2\times10^3}$$

By applying KVL we get,

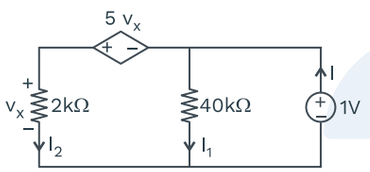
$-v_x + 5v_x = 0$

$v_x = 0$ volt

$I_{SC} = 3A$

Step 2:

Find R_{Th} or R_N



Rack Your Brain

Can we apply Norton's theorem in AC circuits?

Applying KVL we get,

$-v_x + 5v_x + 1 = 0$

$$v_x = -\frac{1}{4}$$

$$I_2 = \frac{v_x}{2k\Omega} = \frac{-1}{8k\Omega}$$

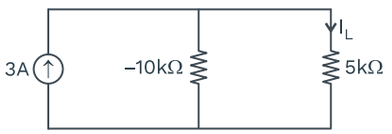
$$I_1 = \frac{1}{40k\Omega}$$

$$I = I_1 + I_2 = \frac{1}{40k\Omega} - \frac{1}{8k\Omega} = \frac{-4}{40k\Omega} = \frac{-1}{10k\Omega}$$

$$R_{Th} = \frac{1}{I} = -10k\Omega \text{ (Practically incorrect)}$$

Note:

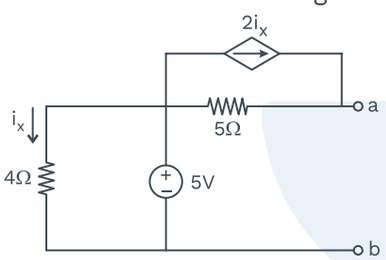
The Thevenin circuit is a simplified form of an electrical circuit, and Thevenin resistance and sign negative doesn't mean that real resistance became negative.



$$I_L = \frac{3 \times -10}{-10 + 5} = 6A$$

Example 8:

Obtain the Norton's resistance (R_N) and Norton's current (I_N) in the given the circuit at terminals a and b using Norton's theorem.

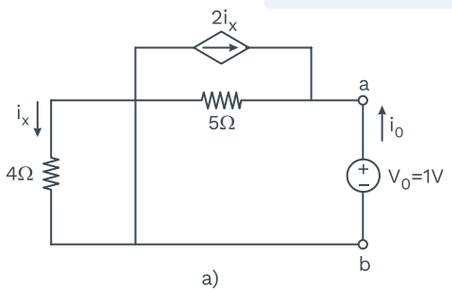


Rack Your Brain

What are the limitations of Norton's theorem?

Sol:

- To obtain Norton's resistance (R_N), the independent voltage source is set to zero.
- Connect the voltage source (V_0 = 1V), to the terminal, and then the circuit obtained is shown in figure a).



a)

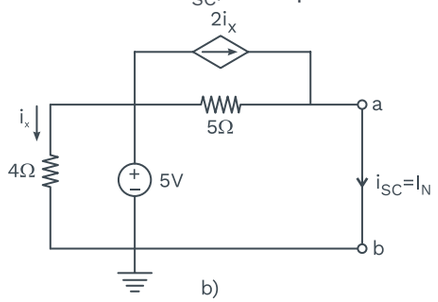
Due to short-circuit, ignore the 4Ω resistor and the 5Ω resistor; the voltage source and the dependent current source will be in parallel. Thus, current i_x = 0A.

Now applying KCL at node A, we get:

$$i_0 = \frac{1V}{5\Omega} = 0.2A$$

$$R_N = \frac{V_0}{i_0} = \frac{1}{0.2} = 5\Omega$$

To obtain Norton's current I_N, terminal a and b will be short-circuited and then find the current I_{SC}, as represented in figure b).



b)

$$i_x = \frac{5}{4} = 1.25A$$

On applying KCL at node a, we get:

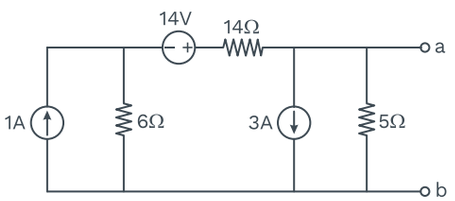
$$\frac{0-5}{5} - 2i_x + I_{SC} = 0$$

$$I_{SC} = 1 + 2\ i_x = 1 + 2 \times 1.25 = 3.5A$$

Thus, $I_N = 3.5A$

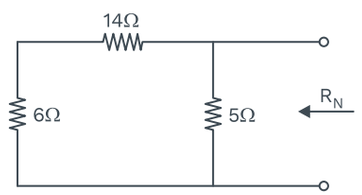
Example 9:

For the circuit shown below, calculate the Norton's equivalent at terminal a and b.



Sol:

To obtain Norton's resistance R_N, the independent sources are set to zero. Thus, the circuit is redrawn as:



a)

Rack Your Brain

Norton theorem is applicable to:

1) Bipolar junction transistor
2) Diodes
3) Resistances
4) Linear networks

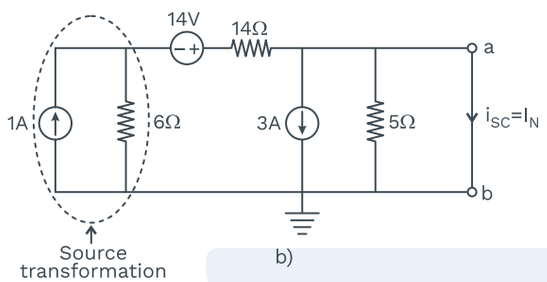
Which of the following options is/are correct:

a) 1 and 3
b) 1, 2 and 3
c) 3 and 4
d) 1, 2 and 4

$R_N = (14 + 6) \parallel 5$

$R_N = 4\Omega$

To obtain Norton's current I_N, the terminal a and b will be short-circuited, as shown in figure b). The 5Ω resistor will not be considered because it has been short-circuited.



Applying nodal analysis at node a, we get:

$$\frac{0-20}{20} + 3 + i_{SC} = 0$$

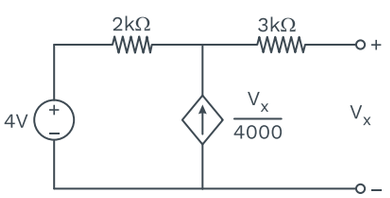
$$-1 + 3 + i_{SC} = 0$$

$$i_{SC} = -2A$$

$$I_N = -2A$$

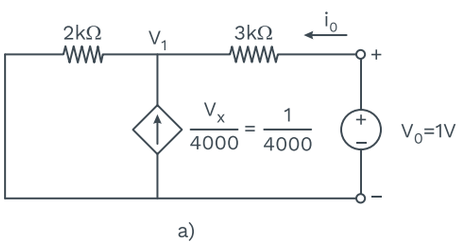
Example 10:

For the circuit shown below, find the Norton equivalent at the terminals.



Sol:

To obtain R_N, the independent voltage source is set to zero. The circuit is redrawn as shown below in figure a).



Applying nodal analysis at V_1, we get:

$$\frac{V_1 - 0}{2000} + \frac{V_1 - 1}{3000} - \frac{1}{4000} = 0$$

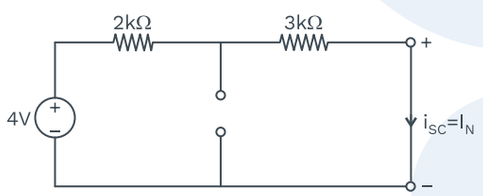
$$\frac{5V_1}{6000} = \frac{7}{12000}$$

$$V_1 = \frac{7}{10} V$$

$$i_0 = \frac{1 - \frac{7}{10}}{3000} = \frac{3}{10 \times 3000} = \frac{1}{10 \times 1000}$$

$$R_N = \frac{V_0}{i_0} = \frac{1}{\frac{1}{10 \times 1000}} = 10k\Omega$$

To find I_N, we short-circuited the terminals, as shown in figure b).
Hence $V_x = 0$



b)

$$i_{SC} = \frac{4}{2k + 3k} = 0.8\,mA$$

Thus, I_N = 0.8 mA

Grey Matter Alert!

Norton's theorem is dual of Thevenin theorem.

MAXIMUM POWER TRANSFER THEOREM (MPT)

- "In any linear, active, bilateral network consisting of a number of energy sources with internal resistances driving an external load, then power transfer to the load is maximum when the load resistance is equal to the source resistance in the circuit".
- If the load is variable, we need to find a condition in which maximum power is absorbed by the load.
- For finding the maximum power transfer, Thevenin's equivalent is very useful. Figure 3.9) demonstrates the power delivered to the load, when the circuit is replaced by Thevenin's equivalent except for the load.

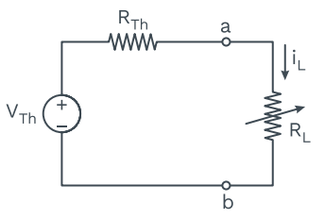


Fig. 3.9 Circuit Used for Maximum Power Transfer

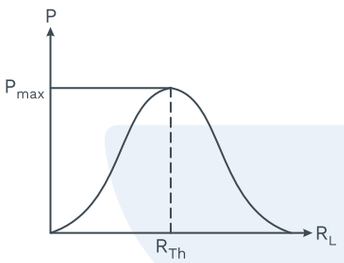


Fig. 3.10 Power Delivered to the Load as a Function of R_L

In the above network, the Thevenin's voltage (V_{Th}) and Thevenin's resistance (R_{Th}) are fixed. The power delivered to the load fluctuates as the load resistance R_L varies, as shown in figure 3.10.

In figure 3.10, for small or large values of load resistance R_L, the power is less and for R_L between 0 to ∞, the power is maximum.

In figure 3.9, if the load is variable, we need to find a condition in which the load absorbs the maximum power.

The conditions for maximum power transfer theorem:

$$I = \frac{V_{Th}}{R_{Th} + R_L}$$

Load power,

$$P_L = |I|^2 R_L$$

$$= \frac{V_{Th}^2 R_L}{\left(R_{Th} + R_L\right)^2}$$

for maximum load power

$$\frac{dP_L}{dR_L} = 0$$

Rack Your Brain

What are the limitations of the maximum power transfer theorem (MPT)?

$$\frac{dP_L}{dR_L} = V_{Th}{}^2\left[\frac{(R_{Th}+R_L)^2 - 2R_L(R_{Th}+R_L)}{(R_{Th}+R_L)^4}\right] = 0$$

$$\frac{V_{Th}^2(R_{Th}+R_L)}{(R_{Th}+R_L)^4}\left[R_{Th}+R_L-2R_L\right] = 0$$

$$R_{Th} - R_L = 0$$

$$R_L = R_{Th}$$

For P_{max}

$$I = \frac{V_{Th}}{R_{Th}+R_L}$$

$$= \frac{V_{Th}}{2R_{Th}}$$

For $P_{max} = I^2R_L$

$$= \frac{V_{Th}^2}{4R_{Th}^2}R_{Th}$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

Efficiency:

Maximum power in load $= \frac{V_{Th}^2}{4R_{Th}}$

Current at maximum power $= \frac{V_{Th}}{2R_{Th}}$

Source power $(P_S) = V_{Th}I$

$$= \frac{V_{Th}^2}{2R_{Th}}$$

$$\text{Efficiency } \eta = \frac{\frac{V_{Th}^2}{4R_{Th}}}{\frac{V_{Th}^2}{2R_{Th}}} \times 100\,\% = 50\,\%$$

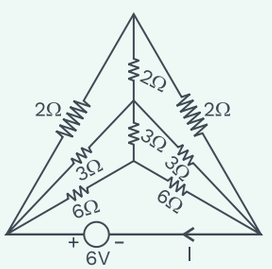
Definitions

Maximum Power Transfer Theorem: "The dc voltage source will deliver maximum power to the variable load resistance only when the load resistance is equal to the source resistance ($R_L = R_{Th}$)"

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

Previous Years' Question

Consider the following circuit:

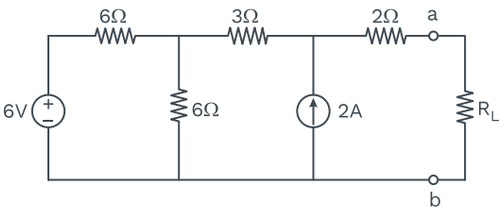


Determine the value of the current (I) in the above circuit ?

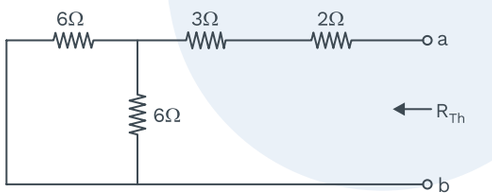
a) 1 A **b)** 2 A

c) 3 A **c)** 4 A

Example 11:
For the circuit shown below, calculate the value of load resistance R_L and also obtain the maximum power.

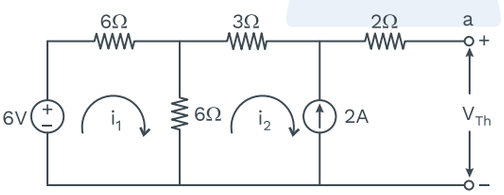


Sol:
To obtain R_{Th} and V_{Th} across the terminal a and b, the circuit can be drawn as-



a)

$R_{Th} = (6||6) + 3 + 2 = 8\Omega$



b)

For V_{Th}, the circuit is shown in figure b).
Now apply mesh analysis in the circuit, we get:

$$-6 + 6i_1 + 6(i_1 - i_2) = 0,\ i_2 = -2A$$

$$12i_1 - 6i_2 = 6$$

$$12i_1 = 6 + 6(-2)$$

$$= -6$$

$$i_1 = -0.5A$$

Now apply the KVL in the outer loop to get V_{Th} across terminal a and b.

Grey Matter Alert!

Applications of the maximum power transfer theorem (MPT) are given as:

- Radio communications
- Automobile engines
- Audio systems

Hence,

$-6 + 6 \times i_1 + 3 \times i_2 + 2(0) + V_{Th} = 0$

$-6 + 6(-0.5) + 3(-2) + 0 + V_{Th} = 0$

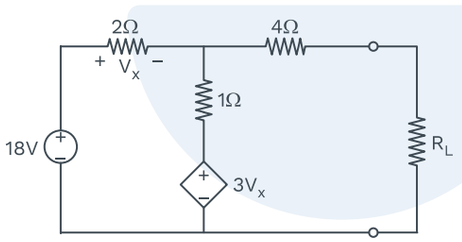
$V_{Th} = 15V$

For maximum power transfer $R_L = R_{Th} = 8\Omega$, and the maximum power transfer

$$P_{max} = \frac{V_{Th}^2}{4R_L} = \frac{15^2}{4 \times 8} = 7.03\,W$$

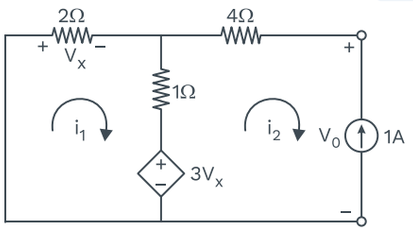
Example 12:

Find the load resistance value in the circuit that will draw the maximum power. Also, determine the maximum power in the circuit.



Sol:

To obtain the Thevenin's resistance R_{Th}, consider the circuit shown in figure a). Now apply mesh analysis, we get:



a)

$2i_1 + 1(i_1 - i_2) + 3V_x = 0,\ V_x = 2i_1$

$i_2 = -1A$

$2i_1 + i_1 - i_2 + 3(2i_1) = 0$

$$i_1 = -\frac{1}{9}A$$

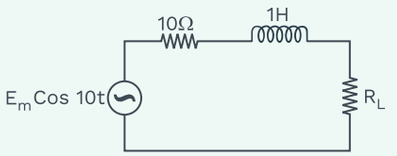
$2i_1 + 4i_2 + V_0 = 0$

$$V_0 = \frac{2}{9} + 4 = \frac{38}{9} = 4.222V$$

$$R_{Th} = \frac{V_0}{i_0} = \frac{4.222}{1} = 4.222\Omega$$

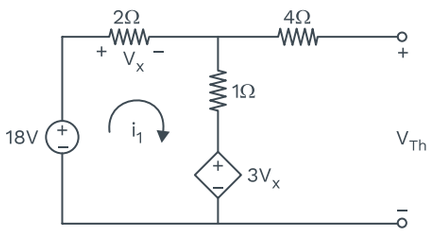
Previous Years' Question

For the figure shown, the value of the load resistor R, that maximizes the power delivered to it is:



a) 14.14 Ω **b)** 10 Ω

c) 200 Ω **d)** 28.28 Ω



b)

To obtain the Thevenin's voltage (V_{Th}), we consider the circuit in figure b).
On applying mesh analysis, we get:

$-18 + 2i_1 + i_1 + 3V_x = 0$, $\quad V_x = 2i_1$

$-18 + 2i_1 + i_1 + 3(2i_1) = 0$

$9i_1 = 18$

$i_1 = 2A$

Now apply KVL in the outer loop, we get:

$-18 + 2i_1 + 4(0) + V_{Th} = 0$

$V_{Th} = 18 - 2 \times i_1$

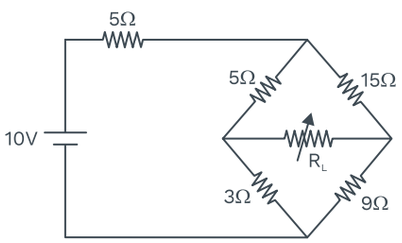
$V_{Th} = 18 - 2 \times 2$

$V_{Th} = 14$ V

Maximum power transfer $P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{14^2}{4 \times 4.222} = 11.6$ W

Example 13:

For the circuit shown below, find the maximum power transferred to R_L.



Sol:

For balance bridge:

$5 \times 9 = 15 \times 3$

$45 = 45$

Under this condition bridge is balanced. Hence no current flow through R_L.

Thus, power dissipation $P_{Diss} = 0$

$$P_{max} = 0$$

Rack Your Brain

For the maximum power to be transferred, the internal resistance or the source resistance must be:

a) greater than the load resistance
b) less than the load resistance
c) equal to the load resistance
d) none of the above

RECIPROCITY THEOREM

"In a linear, passive, bilateral, single-source network, the ratio of the excitation to response is always constant when the positions of the excitation and response are interchanged".

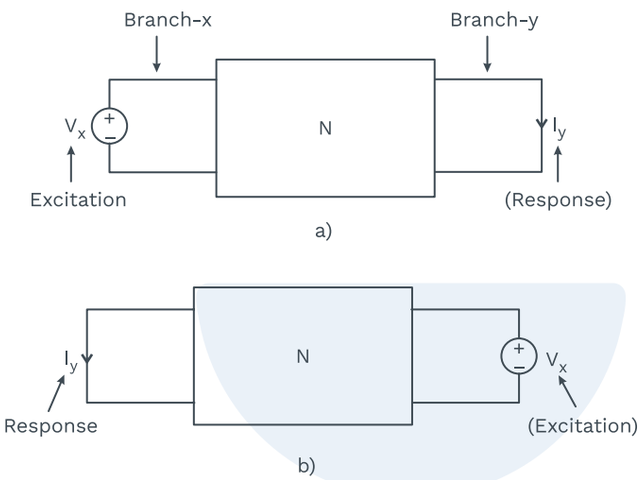


Fig. 3.11 Linear Bilateral Network

$$\frac{V}{I} \text{Constant} \xrightarrow{\text{Apply homogeneity theorem}} \frac{I_1}{V_1} = \frac{I_2}{V_2}$$

- Only one independent source is allowed.
- No limit on the number of dependent sources.
- This theorem is the symmetry of impedance and admittance matrix.

$$Z = \begin{bmatrix} Z_{11} & Z_{12} & Z_{13} \\ Z_{21} & Z_{22} & Z_{23} \\ Z_{31} & Z_{32} & Z_{33} \end{bmatrix}$$

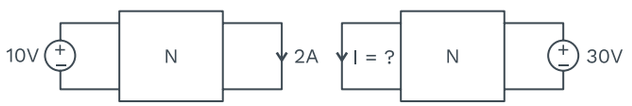
$$Z_{12} = Z_{21},\ Z_{13} = Z_{31},\ Z_{23} = Z_{32}$$

- The reciprocity theorem is always applicable to the reciprocal network.
- For this theorem, the configuration of the network remains unchanged, whereas only the external conditions of the network are changed.
- Circuit should be linear and time-invariant.
- Only one independent source is present in the circuit.
- In the circuit, there must be no dependent source.
- The ratio of response to excitation is either ohm or mho.

Note:

This theorem is applicable only for linear, passive, and bilateral networks, i.e. networks with R, L, C and transformer as elements.

Example 14:
Find the value of current I, when the given network satisfies the reciprocity conditions.



Sol:

$$\left.\frac{\text{Response}}{\text{Excitation}}\right|_1 = \frac{2}{10}$$

$$\left.\frac{\text{Response}}{\text{Excitation}}\right|_2 = \frac{I}{30}$$

According to the reciprocity theorem:

$$\frac{I}{30} = \frac{2}{10}$$

$$I = 6A$$

Rack Your Brain

What are the limitations and applications of the reciprocity theorem?

Example 15:
Use the data of figure a) and calculate the current in the circuit of figure b) is.

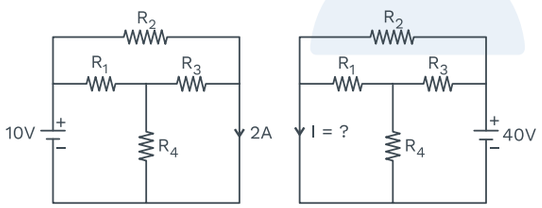


Fig. a) Fig. b)

Sol:

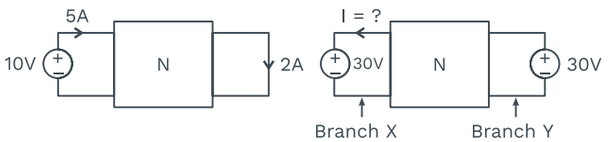
i) All linear circuit
ii) One independent source
iii) No dependent source
iv) Response/Excitation = MHO

$$\left.\frac{\text{Response}}{\text{Excitation}}\right|_1 = \frac{2}{10} \qquad \left.\frac{\text{Response}}{\text{Excitation}}\right|_2 = \frac{I}{40}$$

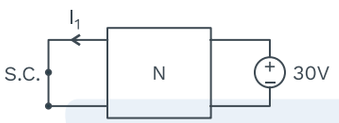
From the reciprocity theorem, $\frac{2}{10} = \frac{I}{40}$

$$I = 8A$$

Example 16:
Find the value of current I when network N is satisfying all the reciprocity conditions.

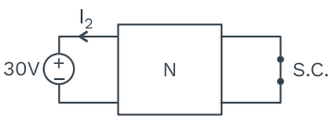


Sol:
Current due to 30-volt voltage source present in branch Y.



From reciprocity theorem $\frac{2}{10} = \frac{I_1}{30} \Rightarrow I_1 = 6A$

Current due to 30-volt voltage source present in branch X.



From reciprocity theorem, $\frac{5}{10} = \frac{-I_2}{30}$

From the superposition theorem, we get:

$I = I_1 + I_2$

$I = 6 - 15 = -9A$

Definitions

Reciprocity Theorem:
"Reciprocity theorem states that in any linear bilateral passive network, the ratio of response to excitation is constant even excitation can be interchanged".

STAR-DELTA TRANSFORMATIONS (WYE-DELTA)

- In the network analysis, the resistances or impedances are connected in various configurations.
- Among these various configurations, the most used configuration is the star-delta or delta-star configuration.
- This configuration is used to simplify the complex circuits in the network theory.
- Figure 3.12) shown below is the bridge circuit. In this circuit, the resistors connected are neither in series nor in parallel, so the combination of resistors R_1 through R_6 would be difficult. So, the below circuit can be simplified by using the 3-terminal equivalent network (i.e. Δ-Y or Y-Δ), is the circuit shown in the figure:

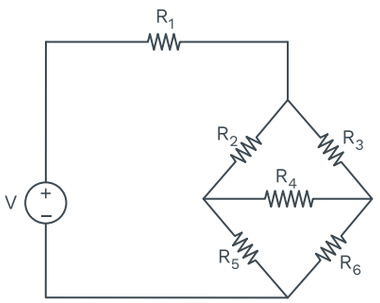


Fig. 3.12 Bridge Network

Rack Your Brain

Is the reciprocity theorem applicable for circuits comprising of dependent sources?

- The circuit shown in figure 3.13 a) is the WYE (Y) network, and the circuit shown in figure 3.13 b) is the tee (T) network.

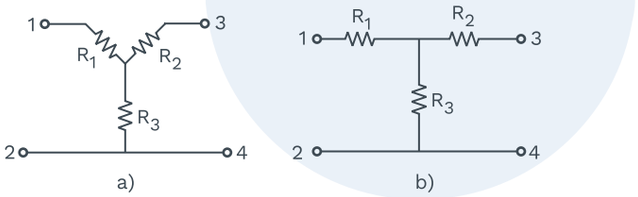


Fig. 3.13 Two Forms of the Same Network a) Y b) T

- The circuit shown in figure 3.14 a) is the delta (Δ) network, and the circuit shown in figure 3.14 b) is the pi Π) network.

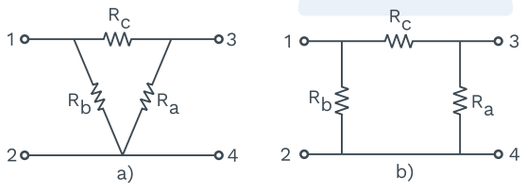


Fig. 3.14 Two Forms of the Same Network: a) Δ b) π

- Figure 3.15 shows both the interconnections superimposed in the same circuit. The formula from wye to delta and delta to wye is used as follows:

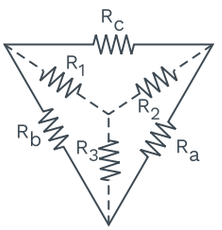


Fig. 3.15 Δ to Y Transformation

Rack Your Brain

What is the benefit of conversion from delta to star and from star to delta?

WYe to delta conversion (Y-Δ):

To convert star network (Y) into equivalent delta (Δ) form, the new values of resistors are calculated as:

$$R_a = \frac{R_1R_2 + R_1R_3 + R_2R_3}{R_1} \quad ...(1)$$

$$R_b = \frac{R_1R_2 + R_1R_3 + R_2R_3}{R_2} \quad ...(2)$$

$$R_c = \frac{R_1R_2 + R_1R_3 + R_2R_3}{R_3} \quad ...(3)$$

Keep in mind:

- The resistor in the delta (Δ) network is the summation of all the possible products of star (Y) resistors touching at a time and divided by the opposite star (Y) resistor.
- The star (Y) and delta (Δ) networks are said to be balanced when:

  $R_1 = R_2 = R_3 = R_Y$ and $R_a = R_b = R_c = R_\Delta$
- Under the given conditions, the conversion formula becomes:

  $$R_Y = \frac{R_\Delta}{3} \text{ or } R_\Delta = 3R_Y \quad ...(4)$$
- One may wonder why R_Y is less than R_Δ. The reason for it is that the Y-connection is like a "series" connection, while the Δ-connection is like a "parallel" connection.

Delta to wye conversion:

To convert Δ-interconnection into equivalent Y-form, the new resistor values are calculated as:

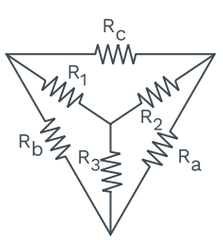


Fig. 3.16 Delta to Wye Conversion

Grey Matter Alert!

The WYE (Y)-delta (Δ) introduced transformation in the year 1899 by Arthur Edwin Kennelly. It is mostly used in the analysis of three-phase electric circuits.

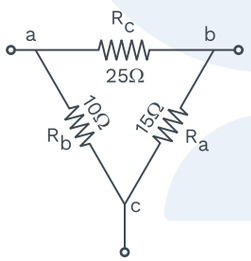
$$R_1 = \frac{R_b R_C}{R_a + R_b + R_c} \quad \ldots(1)$$

$$R_2 = \frac{R_a R_C}{R_a + R_b + R_c} \quad \ldots(2)$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} \quad \ldots(3)$$

Keep in Mind:
The resistor in the star (Y) network is the product of the resistors in two adjacent delta (Δ) branches divided by the summation of the three delta (Δ) resistors.

Example 17:
Convert the given delta (Δ) network in the figure to an equivalent star (Y) network.



Sol:

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$= \frac{10 \times 25}{15 + 10 + 25} = \frac{250}{50} = 5\Omega$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c}$$

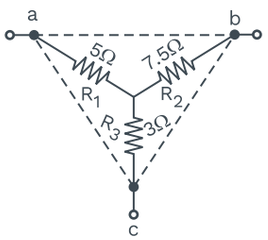
$$= \frac{25 \times 15}{15 + 10 + 25} = \frac{375}{50} = 7.5\Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{15 \times 10}{15 + 10 + 25} = \frac{150}{50} = 3\Omega$$

Grey Matter Alert!

Need for star-delta transformation are:

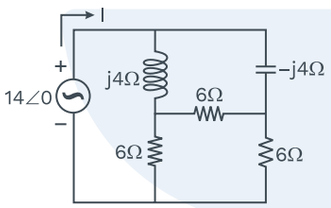
1) Availability of neutral point.
2) Suitable for carrying voltages over long distances.
3) Economic
4) Components require very little space.



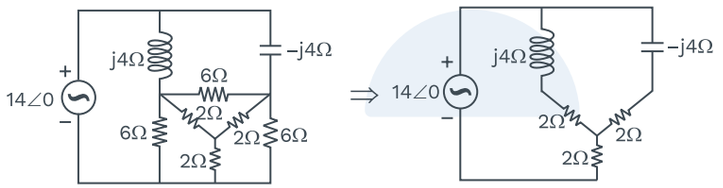
Thus, the equivalent star (Y) network is shown in the above figure.

Sol:

Obtain the equivalent impedance 'Z_{eq}' in the given circuit and calculate the current I.



Solution:



$$Z_{eq} = \left[(2+4j) \,||\, (2-4j)\right] + 2$$

$$Z_{eq} = \left[\frac{(2+4j)(2-j4)}{2+4j+2-j4}\right] + 2$$

$$Z_{eq} = \left[\frac{(2)^2 - (j4)^2}{4}\right] + 2$$

$$Z_{eq} = \left(\frac{4+16}{4}\right) + 2 = \frac{20}{4} + 2 = 5+2 = 7\Omega$$

Current, I:

$$I = \frac{14\angle 0}{7} = 2\angle 0 \text{ A}$$

Hence, the value of current i is $2\angle 0$ A.

Chapter Summary



1) A network is said to be linear if it contains a number of linear elements, linear dependent sources and linear independent sources.

2) Network theorems are used in a network where several sources, nodes and meshes are present, and the response in a single element is desired.

3) In a linear bilateral network, the current or the voltage across an element equals to the algebraic sum of the currents or voltages produced by each source.

4) In a two-terminal network, the procedure of transformation of the voltage source in series with a resistor to an equivalent current source in parallel with the same resistor or vice versa is referred to as source transformation.

5) A linear circuit which contains several voltage sources and resistances can be replaced by a single voltage in series with a resistor using Thevenin's theorem.

6) A linear two-terminal network containing dependent and independent sources can be represented by an equivalent circuit consisting of Norton's equivalent current source (I_N) and Norton's equivalent resistance (R_N) in parallel with it.

- The Thevenin and the Norton theorems are related by a source transformation as:

$$R_N = R_{Th};\ I_N = \frac{V_{Th}}{R_{Th}}$$

7) The power transfer efficiency should be 50% to satisfy the condition of the maximum power transfer theorem in any circuit.

8) The formulas for a delta to wye transformation are:

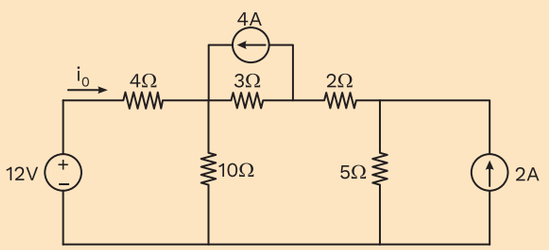
$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c};\ R_2 = \frac{R_c R_a}{R_a + R_b + R_c};\ R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

9) The formulas for wye-to-delta transformation are:

$$R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1};\ R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2};\ R_c = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_3};$$

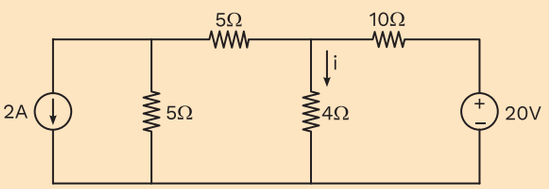
PRACTICE QUESTIONS

Q1 **For the circuit given below, use superposition to obtain i_0.**



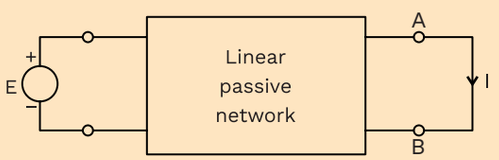
a) 111.11 mA **b) 50 mA** **c) 95 mA** **d) 30.93 mA**

Q2 **For the circuit in the figure, use source transformation to find i.**



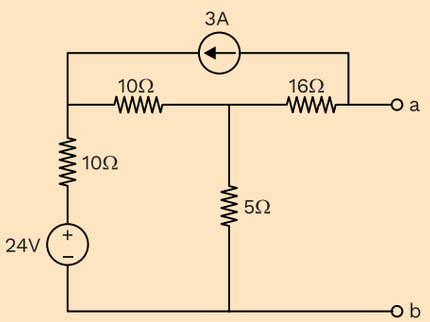
a) 550 mA **b) 555.5 mA** **c) 100 mA** **d) 213.98 mA**

Q3 **For the circuit shown in the figure, when the voltage E is 10V, the current I is 1A. If the applied voltage through the terminals is 90V, the short circuit current flowing through the terminal A–B will be:**



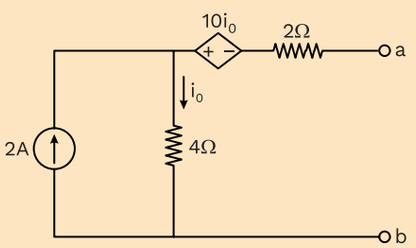
a) 0.9 A **b) 1 A** **c) 9 A** **d) 90 A**

Q4 **Obtain the Thevenin equivalent at the terminal a and b, in the circuit given:**



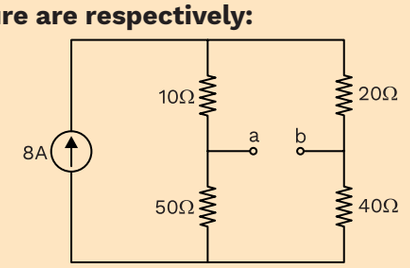
a) V_{Th} = 45V, R_{Th} = 20Ω
b) V_{Th} = 20V, R_{Th} = 3Ω
c) V_{Th} = −49.2V, R_{Th} = 20Ω
d) V_{Th} = 49V, R_{Th} = 30Ω

Q5 **For the circuit shown in the figure, the Norton equivalent resistance (R_N) and Norton equivalent current (I_N) across the terminals a and b are:**



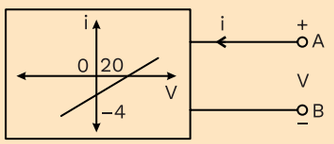
a) R_N = 3Ω, I_N = 3A
b) R_N = 1Ω, I_N = 6A
c) R_N = −4Ω, I_N = 3A
d) R_N = 6Ω, I_N = 18A

Q6 **The Thevenin and the Norton equivalents across the terminals a and b for the circuit shown in the figure are respectively:**



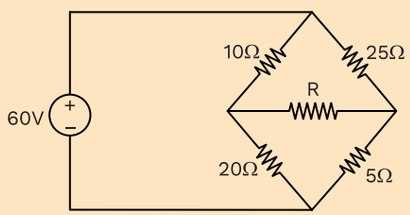
a) V_{Th} = 40V, R_{Th} = R_N = 22.5Ω, I_N = 1.7778A
b) V_{Th} = 40V, R_{Th} = R_N = 20Ω, I_N = 2A
c) V_{Th} = 10V, R_{Th} = R_N = 6Ω, I_N = 5A
d) V_{Th} = 60V, R_{Th} = R_N = 5Ω, I_N = 3.5A

Q7 **A device whose characteristic is shown in the figure below, the resistance seen across the terminals, A and B is:**



a) −5Ω b) $-\frac{1}{5}\Omega$ c) $\frac{1}{5}\Omega$ d) 5Ω

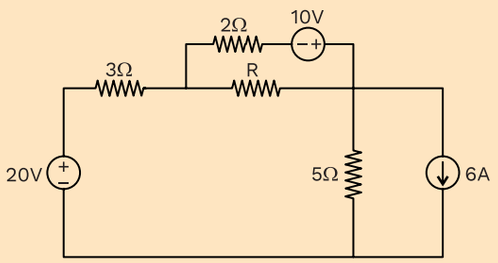
Q8 **The maximum power delivered to the variable resistor R in the circuit given below is:**



a) 30 W b) 20.77 W c) 100.5 W d) 23 W

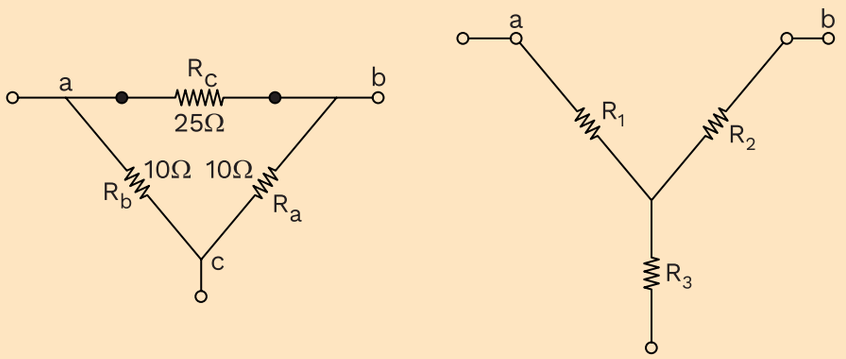
Q9 **Among the following options, which of the following does not apply to the superposition theorem?**
a) Voltage calculation
b) Bilateral elements
c) Power calculations
d) Passive element

Q10 **For the network shown below, the maximum power delivered to the resistor R is:**



a) 1 W **b) 5 W** **c) 0.125 W** **d) 0.625 W**

Q11 **Convert the Δ-network in the figure to an equivalent Y-network.**



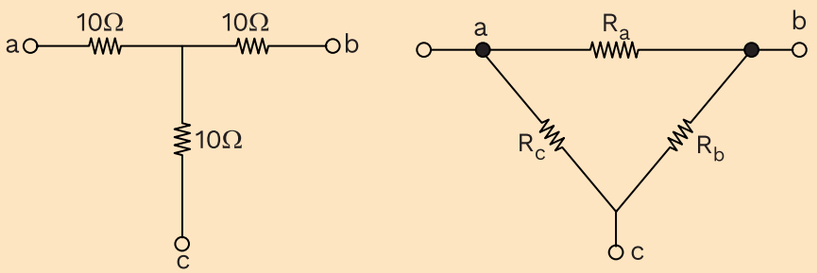
a) $R_1 = 5\Omega, R_2 = 7\Omega, R_3 = 6\Omega$

b) $R_1 = 7\Omega, R_2 = \frac{50}{9}\Omega, R_3 = \frac{50}{9}\Omega$

c) $R_1 = \frac{50}{9}\Omega, R_2 = \frac{50}{9}\Omega, R_3 = \frac{20}{9}\Omega$

d) $R_1 = \frac{50}{9}\Omega, R_2 = \frac{20}{9}\Omega, R_3 = \frac{50}{9}\Omega$

Q12 **Convert the circuits in the figure from Y to Δ.**



a) $R_a = R_b = R_c = 30\Omega$

b) $R_a = R_b = R_c = \frac{10}{3}\Omega$

c) $R_a = R_b = R_c = 10\Omega$

d) $R_a = R_b = R_c = 20\Omega$

Q13 **A resistance R is fed by a network N as shown in figure a), the power consumed by the network is P. If an identical network is added to the network as shown in figure b), then the power consumed by the resistor R is:**

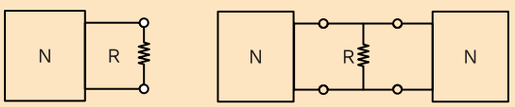


fig. a) fig. b)

a) Less than P

b) Between P and 2P

c) More than P

d) Between P and 4P

Q14 **Calculate the total percentage (%) of the maximum power delivered to the load resistance (R_L) = $3R_{Th}$?**

a) 70

b) 45

c) 65

d) 75

Answer Key:

1)	a)	2)	b)	3)	c)	4)	c)	5)	c)
6)	a)	7)	d)	8)	b)	9)	c)	10)	d)
11)	c)	12)	a)	13)	d)	14)	d)		